Raman Dhiman (FNU Raman, Raman Raman)

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McKetta Department of Chemical Engineering
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The University of Texas at Austin
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EDUCATION

Ph.D. Chemical Engineering, The University of Texas at Austin (expected) May 2028
Dissertation: Biomimetic Membranes for Neuromorphic Computing
Advisors: Manish Kumar, Ph.D. and Benny D. Freeman, Ph.D. | GPA: 4.00/4.00

B.Tech. Chemical Engineering, Indian Institute of Technology Gandhinagar August 2020
GPA: 9.17/10.00 | Institute Silver Medal (second-highest cumulative performance index in the department)

ACADEMIC APPOINTMENTS

Graduate Research Assistant, The University of Texas at Austin 2022–Present
McKetta Department of Chemical Engineering

Graduate Teaching Assistant, The University of Texas at Austin Spring 2023, Fall 2023, Fall 2025
McKetta Department of Chemical Engineering

Junior Research Fellow, Indian Institute of Technology Gandhinagar 2021–2022
Department of Chemical Engineering

RESEARCH EXPERIENCE

Graduate Research Assistant, The University of Texas at Austin August 2022 – Present
McKetta Department of Chemical Engineering | Advisors: Manish Kumar and Benny D. Freeman

- Characterize ion transport, selectivity, and voltage dependence in artificial lipid membranes using droplet interface bilayer electrophysiology, relating pore and peptide structure to conductance behavior.
- Demonstrated the emergence of memristance in jammed interconnected bilayer emulsions using alamethicin, establishing that a 3D-printable tissue mimic reproduces voltage-dependent memory behavior; this work formed part of a Nature Materials publication.
- Showed that blocker molecules induce nonlinear conductance and emergent memristance in open pores that otherwise exhibit ohmic current–voltage behavior, extending memristive dynamics beyond voltage-gated channels (first-author paper in *Faraday Discussions*).
- Developing more stable brain-mimicking bilayer networks and evaluating pore and peptide combinations for neuromorphic computing applications.
- Contributed membrane characterization and electrophysiology to collaborative projects on lanthanide-selective artificial channels, dehydrated biomimetic membranes, and protective carbon fabrics, resulting in *co-authorship* on *four publications*.
- Mentored eight graduate, undergraduate, and intern researchers, four of whom are co-authors on peer-reviewed publications.

This work has resulted in five peer-reviewed research articles (one first-author), one invited review, four published general discussion contributions, and one international patent application, and has been supported by six endowed graduate fellowships at UT Austin.

Junior Research Fellow, Indian Institute of Technology Gandhinagar March 2021 – May 2022
Department of Chemical Engineering | Advisor: Sameer V. Dalvi, Ph.D.

- Micronization and encapsulation of solutes by CO_2-expanded liquid precipitation (PPRGEL) and antisolvent crystallization.

Undergraduate Research Assistant, Indian Institute of Technology Gandhinagar May 2018 – April 2019
Department of Chemical Engineering | Advisor: Kabeer Jasuja, Ph.D.

- Synthesized and characterized boron-based nanosheets and studied their interaction with methylene blue dye in aqueous dispersion.
- Resolved anomalous UV–visible spectra across a concentration matrix of nanosheets and dye, proposing that dye molecules aggregate in the presence of the nanosheets in addition to adsorbing onto them.

This work was presented at the 3rd International Conference on Soft Materials (ICSM 2018).

AWARDS

Outstanding Teaching Assistant Award, McKetta Department of Chemical Engineering, UT Austin 2025–2026

Mulraney Family Endowed Graduate Fellowship in Chemical Engineering 2026–2027

Paige H. and J. Jeff Weidner Endowed Presidential Fellowship in Chemical Engineering 2025–2026

James R. and Merle I. Fair Endowed Graduate Fellowship in Chemical Engineering 2025–2026

Howard A. Halff Endowed Graduate Fellowship in Chemical Engineering 2024–2025

Dr. Thomas F. Edgar Endowed Graduate Fellowship in Chemical Engineering 2022–2023

Sellers Family Endowed Presidential Fellowship in Chemical Engineering 2022–2023

Institute Silver Medal, Indian Institute of Technology Gandhinagar 2020

Awarded to the graduate with the second-highest cumulative performance index in Chemical Engineering, 9th Convocation

Explorer Fellowship, Indian Institute of Technology Gandhinagar 2017

Competitive fellowship supporting a 42-day independent study of regional diversity across India

TEACHING EXPERIENCE

Graduate Teaching Assistant, The University of Texas at Austin Fall 2025
Fluid Flow and Heat Transfer (graduate transport phenomena) | Instructor: Venkat Ganesan, Ph.D.
McKetta Department of Chemical Engineering

Recognized with the departmental **Outstanding Teaching Assistant Award** for this appointment.

Graduate Teaching Assistant, The University of Texas at Austin Fall 2023
Chemical Engineering Process and Project Laboratory (undergraduate) | Instructor: Carlos Landaverde, Ph.D.
McKetta Department of Chemical Engineering

Graduate Teaching Assistant, The University of Texas at Austin Spring 2023
Thermodynamics (undergraduate) | Instructor: Thomas Edison, Ph.D.
McKetta Department of Chemical Engineering

MENTORING EXPERIENCE

Rohit Kantamneni, Undergraduate Student, UT Austin 2026–Present

Benjamin He, Undergraduate Student, UT Austin 2026–Present

Malika Rao, Master's Student, UT Austin 2024–2026
(co-author, Nature Materials and Faraday Discussions)

Kyler Grogan, Undergraduate Student, UT Austin 2023–2026
(current: Engineering Analyst, Kimley-Horn; co-author, Nature Materials)

Kartik Kher, Undergraduate Student, UT Austin 2023–2025
(current: Ph.D. Student, University of Michigan; co-author, ACS Applied Materials & Interfaces)

Lyssa Lashus, Research Intern (High School/Undergraduate), UT Austin 2024, 2025
(co-author, Faraday Discussions)

Rupak Ghosh, Research Intern (High School/Undergraduate), UT Austin 2023, 2026

Angela Wang, Undergraduate Student, UT Austin 2023–2024

PUBLICATIONS

Google Scholar (October 2026): 64 citations | h-index 4 | i10-index 3. Author name shown in bold.

Peer-Reviewed Research Articles

1. Fica, A.; Torbett-Dougherty, M.; West, S.; Rao, M.; **Dhiman, R.**; Wang, J.; Gao, Y.; Tu, Y.-M.; Behera, H.; Roc, C.; Grogan, K.; Beaver, A.; Liu, C.; Lin, A. J.-A.; Belardi, B.; Keitz, B. K.; Hickey, R. J.; Ke, Z.; Rosales, A. M.; Dortdivanlioglu, B.; Sarles, S. A.; Kumar, M. Jammed interconnected bilayer emulsions as 3D-printable biological tissue mimics. *Nature Materials* 2026, 25, 1623–1633. DOI: 10.1038/s41563-026-02679-3

2. **Dhiman, R.**; Gupta, P. R.; Armendarez, N. X.; Lashus, L.; Grogan, K.; Rao, M.; Fica, A.; Vogler, R. J.; Bujanos, D.; Behera, H.; Sarles, S. A.; Kumar, M. Inducing nonlinear conductance and emergent memristance in open pores using blockers. *Faraday Discussions* 2026, 266, 428–445. DOI: 10.1039/d6fd00008h

3. Behera, H.; Duncan, T. J.; Samineni, L.; Oh, H.; Jogdand, A.; Karnik, A.; **Dhiman, R.**; Fica, A.; Hsieh, T.-Y.; Ganesan, V.; Kumar, M. Lanthanide-selective artificial channels. *ACS Nano* 2025, 19 (14), 13927–13940. DOI: 10.1021/acsnano.4c17675

4. Oh, H.; Lopez-Marques, H.; Wamble, N. P.; Vogler, R. J.; **Dhiman, R.**; Behera, H.; Smith, L. A.; Yu, C.; Jogdand, A.; Hsieh, T.-Y.; Hernandez, J.; Mullins, C. B.; Freeman, B. D.; Kumar, M. Highly breathable and protective carbon fabrics. *ACS Applied Materials & Interfaces* 2025, 17 (15), 23383–23393. DOI: 10.1021/acsami.5c01460

5. Oh, H.; Tu, Y.-M.; Samineni, L.; De Respino, S.; Mehrafrooz, B.; Joshi, H.; Massenburg, L.; Lopez-Marques, H.; Elessawy, N.; Song, W.; Behera, H.; **Dhiman, R.**; Boorla, V. S.; Kher, K.; Lin, Y.-C.; Maranas, C.; Aksimentiev, A.; Freeman, B. D.; Kumar, M. Dehydrated biomimetic membranes with skinlike structure and function. *ACS Applied Materials & Interfaces* 2024, 16 (16), 20865–20877. DOI: 10.1021/acsami.3c19572

Review Articles

6. Oh, H.; Samineni, L.; Vogler, R. J.; Yao, C.; Behera, H.; **Dhiman, R.**; Horner, A.; Kumar, M. Approaching ideal selectivity with bioinspired and biomimetic membranes. *ACS Nano* 2025, 19 (1), 31–53. DOI: 10.1021/acsnano.4c09699

Published General Discussion Contributions

7. Iontronics and emergent neuromorphic effects under angstrom-confinement: general discussion. *Faraday Discussions* 2026, 266, 500–529. DOI: 10.1039/d6fd90019d

8. Molecular and ion sieving with angstrom-channels: general discussion. *Faraday Discussions* 2026, 266, 399–410. DOI: 10.1039/d6fd90018f

9. Stimuli-responsive flows in angstrom-scale channels: general discussion. *Faraday Discussions* 2026, 266, 309–332. DOI: 10.1039/d6fd90017h

10. Structure and dynamics of molecules in angstrom-scale confinement: general discussion. *Faraday Discussions* 2026, 266, 169–216. DOI: 10.1039/d6fd90016j

Preprints

11. Torbett-Dougherty, M.; Shaw, T.; Utebay, B.; et al. Capacitance-based measurement of calcium-mediated vesicle fusion upon microinjection in droplet interface bilayers. *ChemRxiv*, August 30, 2026 (preprint). DOI: 10.26434/chemrxiv.15008036/v1

12. Kumar, M.; Gupta, P.; Nance, H.; et al. Explainable AI reveals statistical associations between industrial activity and PFAS contamination of public water systems. *Research Square*, November 6, 2025 (preprint). DOI: 10.21203/rs.3.rs-7935132/v1

PATENTS

Filed

1. Kumar, M.; Vogler, R. J.; Bujanos, D. D.; Kimball, K. E.; **Raman, F.** Methods for the synthesis of nanosheet membranes. International patent application WO 2025/043025 A2, published February 27, 2025.

PRESENTATIONS

Oral Presentations

1. **Dhiman, R.**; Gupta, P. R.; Armendarez, N. X.; Sarles, S. A.; Kumar, M. Inducing nonlinear conductance and emergent memristance in open pores using blockers. Molecular and Ion Flows through Angstrom-Scale Channels, Faraday Discussions, Royal Society of Chemistry. Manchester, United Kingdom. April 13–15, 2026.

Conference Presentations

2. **Dhiman, R.**; James, A. L.; Khandelwal, S.; Dutta, A.; Jasuja, K. Interaction of methylene blue with boron based nanosheets: adsorption and aggregation behaviour. 3rd International Conference on Soft Materials (ICSM 2018), Malaviya National Institute of Technology. Jaipur, India. December 9–14, 2018.

TECHNICAL AND COMPUTATIONAL SKILLS

Electrophysiology: Droplet interface bilayers; jammed interconnected bilayer emulsions; current-voltage characterization; analysis of voltage-dependent ion transport; HEKA and NI instrumentation; Gamry potentiostat; PCB design in KiCad

Membrane and protein methods: Membrane protein expression and purification; cobalt/nickel affinity purification; AKTA/FPLC; lipid vesicle preparation; proteoliposome reconstitution; detergent removal with Bio-Beads; dead-end filtration

Characterization: Dynamic light scattering; scanning and transmission electron microscopy; UV–visible spectroscopy; Raman spectroscopy; optical microscopy; particle size analysis; high-pressure experimentation

Computation and software: MATLAB; Python; experimental trace processing; curve fitting; numerical integration; ODE simulation; molecular dynamics simulation; scientific plotting; COMSOL Multiphysics; ASPEN Plus; ANSYS Fluent; Simulink; Autodesk Inventor

ACADEMIC SERVICE

Coordinator, Tinkerers' Laboratory, Indian Institute of Technology Gandhinagar 2017–2018
Member of the founding team that established an open student prototyping and innovation facility

Senator, Student Senate, Indian Institute of Technology Gandhinagar 2016–2017
Represented the batch to the university administration on academic and student-life matters

Student Reviewer, Earn While You Learn Program, Library Services, IIT Gandhinagar 2019
Reviewed approximately 100 articles from AIChE's Chemical Engineering Progress and recommended selections for the undergraduate collection

COMMUNITY SERVICE AND OUTREACH

Peer Assisted Learning (PAL) Mentor, Indian Institute of Technology Gandhinagar 2017–2018
Mentored two first-year undergraduates adapting to English as the primary medium of instruction

PROFESSIONAL EXPERIENCE

Management Trainee, Dai-ichi Karkaria Limited September 2020 – February 2021
Production and Pilot Plant, Bharuch, Gujarat, India

Rotated through production and pilot plant operations at a specialty chemicals manufacturer, developing a management-level understanding of small-scale process industry operations.